

Topic 9

RRH (2017): Model

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 - ▶ Quantitative Spatial Economics lecture in Princeton University course EC552—by Esteban Rossi-Hansberg

Introduction

Quantitative Spatial Models of Economic Geography (henceforth QSMs)

- ▶ Have discussed **quantitative spatial models**
 - ▶ **Rich** enough to incorporate first-order features of the data
 - ▶ Sufficiently **tractable** as to enable quantitative counterfactuals
 - ▶ Account for interactions on goods and factor markets in **general equilibrium**
- ▶ One distinctive feature is that these models account for **spatial interactions**
 - ▶ So far, we have focused on frictional **mobility of people**
 - ▶ Between cities (Topic 5) and within cities (Topics 6-8)
- ▶ Now we turn to frictional **trade of goods**

Q: At what spatial scale do trade costs matter?

History of thought I

- ▶ Theoretical research on **New Economic Geography** (NEG) concentrated on:
 - ▶ Formalizing mechanisms for **agglomeration and cumulative causation**
 - ▶ Fujita, et al. 1999, Fujita and Thisse 2002 and Baldwin et al. 2003
- ▶ **Agglomeration force** generated by combination of
 - ▶ **love of variety, increasing returns to scale, low (but positive) transport costs**
 - ▶ **Greater market access $\Rightarrow$ lower prices $\Rightarrow$ larger market**
- ▶ Explains emergence of an uneven distribution of economic activity
 - ▶ even on a featureless plain of *ex ante* identical locations
 - ▶ Role for '**historic accident & cumulative causation**'
 - ▶ **Multiple equilibria** in location choices
 - ▶ Recommended enjoyable (no joke!) reading: Davis & Weinstein (2002)
 - ▶ Notice that QSMs are not intended to feature multiple equilibria

History of thought II

- ▶ Theoretical literature stimulated a wave of **empirical research**
 - ▶ Mostly **reduced-form** in nature
 - ▶ Mapping from the model to the empirical specification was often unclear
 - ▶ Often, the focus was on testing stylized predictions
- ▶ Challenges and limitations
 - ▶ Are coefficients of reduced-form relationships invariant to policy intervention?
 - ▶ Example: $\ln L_i^F = \alpha + \beta \ln MA_i + \varepsilon_i$
 - ▶ Transport improvement increases market access
 - ▶ Firms, workers, wages and prices adjust in general equilibrium
 - ▶ Estimated β not necessarily the same in new equilibrium
 - ▶ A spatial version of the **Lucas Critique**
 - ▶ Inability to evaluate **welfare effects** of policy

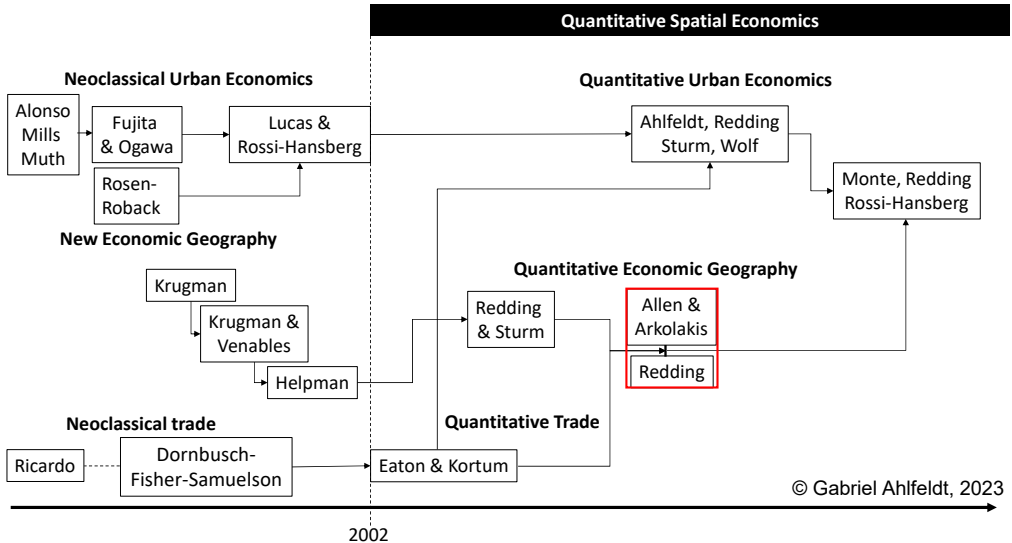
History of thought III

- ▶ QSMs of Economic Geography **connect more closely to observed data**
 - ▶ Following the introduction of **quantitative models of international trade**
 - ▶ In particular, Eaton and Kortum (2002)
 - ▶ Does **not aim at explaining agglomeration of economic activity**
 - ▶ Agglomeration in these models is the **result of exogenous fundamentals**
 - ▶ **Augmented** by endogenous economic mechanisms
 - ▶ These frameworks can accommodate many realistic spatial distributions
 - ▶ Capturing productivity and amenity
 - ▶ The same quantitative framework can be derived from an entire class of theoretical models of economic geography

History of thought IV

- ▶ In many respects **similar to those of QSMs of internal city structure**
 - ▶ Endogenous (spatial) mechanisms focus on **trade instead of commuting**
- ▶ Provide microfoundations for central features of the data
 - ▶ Through accommodating **many regions and a rich geography** of trade costs
- ▶ **Explain the observed data as an equilibrium of the model**
 - ▶ Models are typically exactly identified
 - ▶ **Can be inverted** to identify the unique values of the structural fundamentals
 - ▶ **one-to-one mapping** from the **observed data to fundamentals for given parameters**
 - ▶ Observed variation in the data can be decomposed within the model
 - ▶ into the contributions of each fundamental
 - ▶ "Well-behaved" counterfactuals

History of thought V



Roadmap

- ▶ **Redding-Rossi-Hansberg (2017) model**
 - ▶ A canonical QSM of Economic Geography: Multi-region version of Helpman (1998)
 - ▶ Used in Redding & Sturm (2008)
- ▶ **Topic 9 (today)**
 - ▶ Model
 - ▶ Equilibrium
 - ▶ Quantification
- ▶ **Topic 10**
 - ▶ Counterfactuals (Topic 10)

Building blocks of QSMs

A Menu of Quantitative Spatial Models

- ▶ Taking stocks
 - ▶ QSMs implicitly or explicitly make assumptions about several **building blocks**:
1. Preferences
 2. Production Technology
 3. Technology for Trading Goods
 4. Technology for Idea Flows
 5. Technology for the movement of people
 6. Endowments
 7. Equilibrium

A Menu of QSM: Preferences

- ▶ Homogeneous versus differentiated goods (love of variety)
- ▶ Single versus multiple sectors
- ▶ Exogenous (e.g., scenic views) and/or endogenous amenities (e.g., crime)
- ▶ Fixed local factors in utility (residential land use)
- ▶ Common versus idiosyncratic preferences
 - ▶ Idiosyncratic preferences for each location that are typically modelled as being drawn from an extreme value distribution

What did ARSW (2015) pick?

A Menu of QSM: Production Technology

- ▶ Constant versus increasing returns
- ▶ Exogenous (e.g., mineral resources) and/or endogenous productivity differences (e.g., knowledge spillovers)
 - ▶ Typically need exogenous differences across locations to rationalize the observed employment and income data
- ▶ Input-output linkages
- ▶ Fixed local factors in production (commercial land use)

What did ARSW (2015) pick?

A Menu of QSM: Technology for Trading Goods

- ▶ Variable versus fixed trade costs
 - ▶ Iceberg variable transport costs popular: $d_{ni} \geq 1$ units of a good must be shipped from location i to location n in order for one unit to arrive
 - ▶ Important to be consistent with the gravity equation (bilateral trade increases with exporter and importer size and declines with distance)
- ▶ Asymmetric versus symmetric transport costs
- ▶ Geographic (e.g., mountains) vs. economic frictions (e.g. borders, infrastructure)
- ▶ Non-traded goods

What did ARSW (2015) pick?

A Menu of QSM: Technology for Idea Flows

- ▶ Knowledge externalities and diffusion (agglomeration economies, again)
 - ▶ Marshall (1920) distinguishes between knowledge spillovers, externalities due to thick labor markets, and backward and forward linkages
 - ▶ Duranton & Puga (2004) distinguish between searching, learning, and matching
- ▶ Innovation
 - ▶ Local productivity can be the result of intentional investments in innovation
 - ▶ The incentives depend on the market size / the distribution of economic activity
- ▶ Transferability of ideas
 - ▶ Extent to which ideas can be costlessly transferred to other locations

What did ARSW (2015) pick?

A Menu of QSM: Technology for the movement of people

- ▶ Migration costs
 - ▶ Can be static relative to birthplace or dynamic relative to last location
- ▶ Commuting and commuting costs
 - ▶ Whether agents can separate their workplace and residence by commuting
 - ▶ Standard in urban but not in regional models
 - ▶ Heterogeneity
 - ▶ E.g. across workers of different skill types
- ▶ Congestion in transportation
 - ▶ Transportation costs become endogenous

What did ARSW (2015) pick?

A Menu of QSM: Endowments

- ▶ Population
 - ▶ Groups, e.g. by skill
- ▶ Spatial scope and units
 - ▶ Mostly, geographically mobile labor is combined with geographically immobile land
- ▶ Capital and infrastructure
 - ▶ Other mobile factors of production can be introduced
 - ▶ E.g. physical capital that is used in a construction sector
 - ▶ Durability introduces a dynamic forward-looking problem
 - ▶ Local capital investments over time that do not depreciate fully
 - ▶ The whole distribution of capital across space is the state variable

What did ARSW (2015) pick?

Criteria for Menu Choice

▶ Tractability

- ▶ Results on the existence and uniqueness of equilibrium and for comparative statics
- ▶ Computing power $\Rightarrow$ can solve large systems of non-linear equations numerically

▶ Structural assumptions

- ▶ Determine what is assumed to be a structural parameter or fundamental
- ▶ Plausibly exogenous and invariant to policy interventions?

▶ Connection between model and **data**

- ▶ What are the spatial units for which data are available?
- ▶ Do we observe flows? Else, need to parameterize trade (or commuting) costs
- ▶ **Over-identification** checks
 - ▶ Do we want to retain data to test the model's predictive power?

Redding & Rossi-Hansberg (2017) model

RRH (2017) selection of building blocks

1. **Preferences:** Love of variety; Single traded sector; No amenities; Residential land use; Common preferences
2. **Production technology:** Increasing returns to scale; Exogenous productivity; No input-output linkages; No commercial land use
3. **Technology for the Movement of Goods:** Iceberg variable trade costs; Symmetric trade costs; Economic and Geographic Frictions; No non-traded goods besides residential land use
4. **Technology for the Movement of Ideas:** No knowledge externalities or diffusion; No innovation; No transferability of ideas
5. **Technology for the Movement of People:** Perfectly costless migration; No commuting; Single worker type with no heterogeneity; No congestion in transportation
6. **Endowments:** Homogenous labor; Exogenous land endowments in regions within a single country; No capital
7. **Equilibrium:** Monopolistic competition; General equilibrium with a single country; Land rents redistributed to residents; Trade is balanced in each location

A Canonical Quantitative Economic Geography Model

Important: You can find many of the derivations related to CES demand and monopolistic competition in Allen & Arkolakis' lecture notes (lecture #1) [website](#)

- ▶ Consider an economy consisting of a set N of **regions indexed by n**
 - ▶ Each region is endowed with an **exogenous quality-adjusted supply of land** (H_i)
- ▶ Economy is endowed with a **measure $\bar{L}$ of workers** who supply one unit of labor
- ▶ Workers are **perfectly geographically mobile**
 - ▶ In equilibrium **real wages are equalized** across all populated regions
- ▶ Regions are connected by a **bilateral transport network**
 - ▶ Can be used to **ship goods subject to symmetric iceberg trade costs**
 - ▶ $d_{ni} = d_{in} > 1$ units must be shipped from region i in order for one unit to arrive in region $n \neq i$, where $d_{nn} = 1$

Consumer Preferences

- ▶ Preferences are defined over goods consumption C_n and residential land use h_n

$$U_n = \left(\frac{C_n}{\alpha} \right)^\alpha \left(\frac{h_n}{1-\alpha} \right)^{1-\alpha}, \quad 0 < \alpha < 1$$

- ▶ where $C_n = \left[\sum_{i \in N} \int_0^{M_i} c_{ni}(j)^\rho dj \right]^{\frac{1}{\rho}}$. Given the CES form, F.O.C. deliver:
- ▶ Dual price index $P_n = \left[\sum_{i \in N} \int_0^{M_i} p_{ni}(j)^{1-\sigma} dj \right]^{\frac{1}{1-\sigma}}$
- ▶ Intuitively
 - ▶ Each location $i \in N$ produces a measure of M_i varieties indexed by j
 - ▶ $c_{ni}(j)$ is consumption (at n) of each of these varieties j produced in i
 - ▶ Total consumption is the **aggregate over all varieties sourced from all regions**
 - ▶ The integral aggregates over varieties within origins; the sum aggregates across origins

Production

- ▶ Varieties are produced under conditions of
 - ▶ **monopolistic competition**
 - ▶ downward-sloping demand curves due to horizontally differentiated varieties
 - ▶ Setting $MC = MR$ results in markup $P > MC$
 - ▶ **increasing returns to scale**; implied production function is

$$x_i(j) = A_i (l_i(j) - F)$$

- ▶ where a fixed cost F is paid in terms of units of labour $l_i(j)$
 - ▶ A_i is a location-specific productivity shifter
 - ▶ Hence, to produce $x_i(j)$ units of a variety j in location i , we require:

$$l_i(j) = F + \frac{x_i(j)}{A_i},$$

Production

- ▶ Profit maximization implies $MC = MR \Rightarrow p_{ni}(j) = \underbrace{\left(\frac{\sigma}{\sigma - 1}\right)}_{\text{Markup}} \underbrace{\frac{w_i}{A_i} d_{ni}}_{\text{Marginal Cost}}$

- ▶ Recall that $\sigma = -\frac{\partial c}{\partial p} \frac{p}{c}$ and use it in $MR = \frac{\partial p}{\partial x} x + p$ to get $p = \frac{\sigma}{\sigma - 1} MC$

- ▶ **Zero profits** imply that the equilibrium output of each variety is

$$x_i(j) = \bar{x}_i = A_i(\sigma - 1)F \Rightarrow l_i(j) = \bar{l} = \sigma F$$

- ▶ We set revenue equal to labour cost inflated by trade cost: $p_{ni}(j)x_{ni}(j) = l_i(j)w_id_{ni}$
 - ▶ Plug in p_{ni} and l_i from above and solve for $\bar{x}$; then plug the solution into $l_i(j)$ equation

- ▶ Labour market clearing implies that

$$L_i = \bar{l}M_i \Rightarrow M_i = L_i/\sigma F$$

- ▶ Each region produces the **same amount** of each variety using the same labour input
 - ▶ Regions with **greater employment produce more varieties**

Price Indices and Expenditure Shares

- ▶ The price index can be expressed as $P_n = \frac{\sigma}{\sigma-1} \left(\frac{1}{\sigma F} \right)^{\frac{1}{1-\sigma}} \left[\sum_{i \in N} L_i \left(d_{ni} \frac{w_i}{A_i} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$
 - ▶ Just use $p_{ni}(j)$ and M_j in the dual price index
- ▶ The share of location n 's expenditure on goods produced in location i is

$$\pi_{ni} = \frac{M_i p_{ni}^{1-\sigma}}{\sum_{k \in N} M_k p_{nk}^{1-\sigma}} = \frac{L_i \left(d_{ni} \frac{w_i}{A_i} \right)^{1-\sigma}}{\sum_{k \in N} L_k \left(d_{nk} \frac{w_k}{A_k} \right)^{1-\sigma}}$$

- ▶ Since there are M_i varieties produced at i , n imports value $P_{ni} x_{ni} M_i$
 - ▶ Recall that $-\sigma$ is the price elasticity of demand $\Rightarrow p_{ni}^{1-\sigma}$ captures $p_{ni} x_{ni}$ up to scale
- ▶ **The model implies a „gravity equation” for goods trade.** For $\sigma > 1$, imports
 - ▶ **decrease in trade costs d_{ni} and origin wages**
 - ▶ **increase in productivity and total employment** (implies more varieties)
- ▶ Can use gravity multilateral resistance to express $P_n = \frac{\sigma}{\sigma-1} \left(\frac{L_n}{\sigma F \pi_{nn}} \right)^{\frac{1}{1-\sigma}} \frac{w_n}{A_n}$

Income and Population Mobility

- ▶ **Expenditure on land** in each location is **redistributed** lump sum to local workers
 - ▶ Total income is labor income plus expenditure on housing (trade balance)

$$v_n L_n = w_n L_n + (1 - \alpha) v_n L_n = \frac{w_n L_n}{\alpha}$$

- ▶ Marshallian demand (f.o.c.) delivers $r_n = \frac{(1-\alpha)v_n L_n}{H_n} = \frac{1-\alpha}{\alpha} \frac{w_n L_n}{H_n}$
 - ▶ with H_n being an exogenous endowment (eq. incorporates land market clearing)
- ▶ Population mobility implies uniform real income and indirect utility

$$\bar{V}_n = \frac{v_n}{p_n^\alpha r_n^{1-\alpha}} = \bar{V}$$

- ▶ **Workers get the rent back, but they still live in smaller flats if r is higher**

Income and Population Mobility

- Plug in $\{v_n, P_n, r_n\}$ in spatial equilibrium condition (constant utility)

$$\bar{V} = \frac{A_n^\alpha H_n^{1-\alpha} \pi_{nn}^{-\frac{\alpha}{\sigma-1}} L_n^{\frac{\sigma(1-\alpha)-1}{\sigma-1}}}{\alpha \left(\frac{\sigma}{\sigma-1}\right)^\alpha \left(\frac{1}{\sigma F}\right)^{\frac{\alpha}{1-\sigma}} \left(\frac{1-\alpha}{\alpha}\right)^{1-\alpha}}$$

- Can isolate L_n and compute population share $\lambda_n \equiv L_n/\bar{L}$:

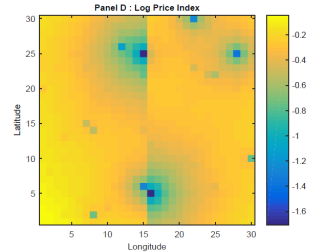
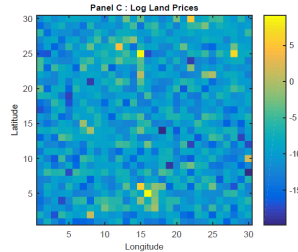
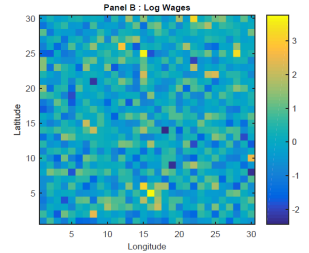
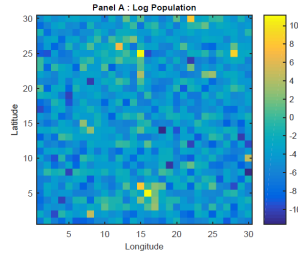
$$\lambda_n = \frac{L_n}{\bar{L}} = \frac{\left[A_n^\alpha H_n^{1-\alpha} \pi_{nn}^{-\alpha/(\sigma-1)} \right]^{\frac{\sigma-1}{\sigma(1-\alpha)-1}}}{\sum_{k \in N} \left[A_k^\alpha H_k^{1-\alpha} \pi_{kk}^{-\alpha/(\sigma-1)} \right]^{\frac{\sigma-1}{\sigma(1-\alpha)-1}}}$$

- Each location's domestic trade share π_{nn} summarizes its **market access** (MA)
 - **High MA** to other locations $\Rightarrow$ low $\pi_{nn} \Rightarrow$ **Greater λ_n due to lower P_n**

Market access (RRH A_n)

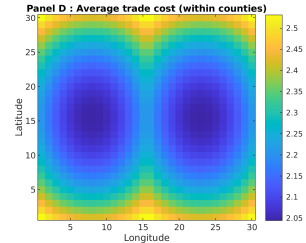
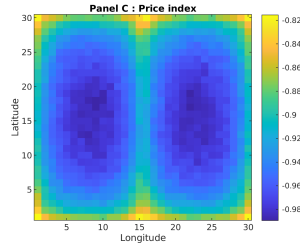
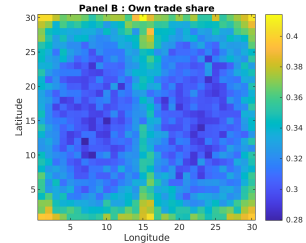
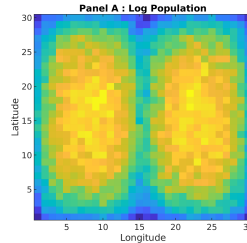
Market access should lead to lower own trade share and a **larger population!**

Ok, but **why don't we see anything?**



Market access (GA A_n)

- ▶ Now with a slightly different parameterization for A_n
- ▶ Central locations have lower average trade costs (within countries)
- ▶ Results in more trade (lower own share) and a lower price index
- ▶ And, indeed, a greater population
 - ▶ Obviously, rents (fixed housing supply) and wages (constant reservation utility) follow the same pattern



Market access (GA A_n)



(a) L_n vs. π_{nn}



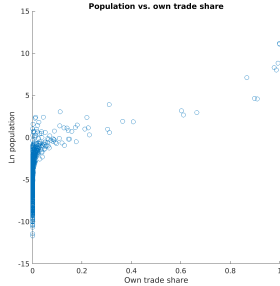
(b) P_n vs. π_{nn}



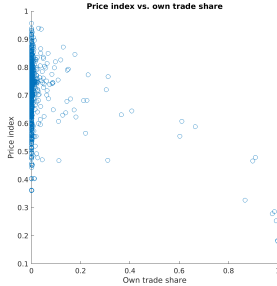
(c) π_{nn} vs. Av. trade cost

Pop. declines in π_{nn} as predicted by Eq. (15) (driven by P_n and trade costs) ✓

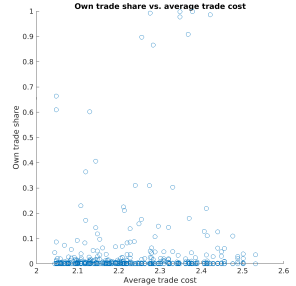
Market access (RRH A_n)



(a) L_n vs. π_{nn}



(b) P_n vs. π_{nn}



(c) π_{nn} vs. Av. trade cost

With RRH productivities (A_n), the patterns reverse! What is going on?

Role of structural residuals

- ▶ **Much greater** variation in A_n chosen by RRH:

- ▶ RRH: `a=normrnd(0,1,NN,1); a=exp(a);`
- ▶ GA: `a=normrnd(0,0.01,NN,1); a=exp(a);`

- ▶ Log-linearize location choice probability Eq. (15):

$$\ln L_n = c - \left[\frac{\alpha}{\sigma(1-\alpha)-1} \right] \ln \pi_{nn} + \underbrace{\left[\frac{\sigma-1}{\sigma(1-\alpha)-1} (\alpha \ln A_n + (1-\alpha) \ln H_n) \right]}_{\text{structural residual}}$$

- ▶ As evident from trade share Eq. (9), $\text{cov}(A_n, \phi_{nn}) > 0$
- ▶ Since $\text{cov}(\pi_{nn}, A_n) > 0$, OVB from A_n on estimated $\frac{\partial L_n}{\partial \pi_{nn}}$ is positive
 - ▶ More variation in $A_n \Rightarrow$ Greater OVB

- ▶ Broader insight applicable to many contexts in structural work

- ▶ If we wanted to estimate α or σ using $\{L_n, \pi_{nn}\}$, we would **need an instrument**

Equilibrium

General Equilibrium

- Wages are implicitly determined by

$$w_n^{1-2\sigma} A_n^{\sigma-1} L_n^{(\sigma-1)\frac{1-\alpha}{\alpha}} H_n^{-(\sigma-1)\frac{1-\alpha}{\alpha}} = \xi$$

- where ξ is a scalar that normalizes wages
 - To get this, plug rent and price index into the spatial equilibrium condition ($\bar{V}$)
 - Notice that since $\{A_n, H_n, \sigma, \alpha\}$ are primitives, L_n maps into w_n (up to scale)
- Follow the mighty Allen & Arkolakis (2014) and substitute wage equation and own trade share (π_{nn}) into location choice probability (λ_n) to get to get

$$L_n^{\tilde{\sigma}\gamma_1} A_n^{\frac{(\sigma-1)^2}{2\sigma-1}} H_n^{-\frac{\sigma(\sigma-1)(1-\alpha)}{\alpha(2\sigma-1)}} = \bar{W}^{1-\sigma} \sum_{i \in N} \frac{1}{\sigma F} \left(\frac{\sigma}{\sigma-1} d_{ni} \right)^{1-\sigma} \left(L_i^{\tilde{\sigma}\gamma_1} \right)^{\frac{\gamma_2}{\gamma_1}} A_i^{\frac{\sigma(\sigma-1)}{2\sigma-1}} H_i^{-\frac{(\sigma-1)^2(1-\alpha)}{\alpha(2\sigma-1)}},$$

- where $\tilde{\sigma} = \frac{\sigma-1}{2\sigma-1}$, $\gamma_1 = \frac{\sigma(1-\alpha)}{\alpha}$, $\gamma_2 = 1 + \frac{\sigma}{\sigma-1} - \frac{(\sigma-1)(1-\alpha)}{\alpha}$
 - the scalar $\bar{W}$ is determined by labor market clearing $\sum_{n \in N} L_n = \bar{L}$ and

General Equilibrium

- ▶ The equilibrium condition uses
 - ▶ the following equations: Trade share (9), price index (8), population mobility (13)
 - ▶ the assumption that trade costs are symmetric ($d_{ni} = d_{in}$)
- ▶ The **equilibrium is referenced solely by L_n**
 - ▶ Can solve for L_n from Eq. (16) for given values of exogenous objects
 - ▶ structural parameters $\{\alpha, \sigma\}$, fundamentals $\{A_n, H_n\}$, trade costs d_{ni} , fixed cost, F
 - ▶ We do not need to solve for any other endogenous variable simultaneously
 - ▶ there is a **recursive structure** to solve for the other endogenous objects
- ▶ **Cannot solve for L_n analytically**
 - ▶ Need to write a solver that deals with this fixed-point problem

Existence and Uniqueness

- ▶ Allen & Arkolakis (2014) show that there is a **unique** solution for L_n as long $\frac{\gamma_2}{\gamma_1} \in (0, 1]$
 - ▶ Solution can be found by **iteration** from any initial distribution of populations
- ▶ Reason is that **congestion forces always dominate agglomeration forces**
 - ▶ Attractive location $\Rightarrow$ population grows $\Rightarrow$ rent increases $\Rightarrow$ attractiveness falls
- ▶ A sufficient condition for $\frac{\gamma_2}{\gamma_1} \in (0, 1)$ is $\sigma(1 - \alpha) > 1$
 - ▶ The higher the elasticity of substitution (σ), the weaker the agglomeration force
 - ▶ The higher the share of land ($1 - \alpha$), the stronger the dispersion force
- ▶ Important property that ensures **determinate solutions for counterfactuals**
 - ▶ Absence of multiple equilibria can also be **frustrating from a policy perspective**

Quantification

Estimation

► Housing expenditure share

- Consensus value (also used by RHH) is $\alpha = 0.25$ (Davis & Ortalo-Magné, 2011)
 - Share in Germany probably closer to $\alpha = 0.33$ (Ahlfeldt & Pietrostefani, 2019)

► Elasticity of substitution

- Consensus elasticity of trade flow with respect to trade cost is 4
 - See Simonovska and Waugh (2014) for an extensive review
- In the RRH model, this elasticity corresponds to $1 - \sigma$
 - See gravity equation for trade shares π_{ni}
- Thus, in this model, the value consistent with empirical estimates is $\sigma = 5$

► Distance elasticity of trade cost

- If we do not observe flows, we need to parameterize trade cost
- RRH use $d_{ni} = dist_{ni}^{\Phi}$, calibrating $\Phi = 0.375$ to match $(\sigma - 1)\Phi = 1.5$
- $(\sigma - 1)\Phi$ is context-dependent (Wrona & Mori, 2024) $\Rightarrow$ **worth estimating**

Distance elasticity of trade flow

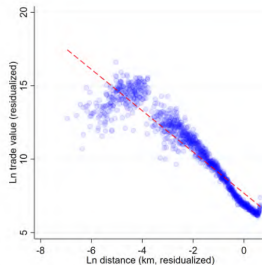
- ▶ **Model-derived gravity equation:**
$$\pi_{ni} = \frac{L_i \left(d_{ni} \frac{w_i}{A_i} \right)^{1-\sigma}}{\sum_{k \in N} L_k \left(d_{nk} \frac{w_k}{A_k} \right)^{1-\sigma}}$$
- ▶ For estimation, we **parameterize** $d_{ni} = \text{dist}_{ni}^{\Phi} \times e_{ni}$, where e_{ni} is a mean-one error
 - ▶ Could also specify in semi-log form: $d_{ni} = \exp(\varphi \text{dist}_{ni} + e_{ni})$
- ▶ **Plugging in d_{ni} , and take logs** delivers the **reduced-form specification**:

$$\ln \pi_{ni} = [(1 - \sigma)\Phi] \ln \text{dist}_{ni} + O_i + D_n + [(1 - \sigma) \ln e_i]$$

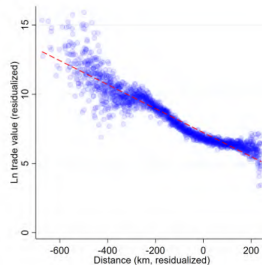
- ▶ In semi-log form this becomes: $\ln \pi_{ni} = [(1 - \sigma)\varphi] \text{dist}_{ni} + O_i + D_n + [(1 - \sigma)e_i]$
 - ▶ $[(1 - \sigma)\Phi]$ or $[(1 - \sigma)\varphi]$ is the distance elasticity of trade flows to be estimated
 - ▶ Origin FE, O_i capture variate in $\{L_i, w_i, A_i\}$ at trade origin
 - ▶ Destination FE, D_n capture multilateral resistance at trade destination
- ▶ See Head & Mayer (2014) for estimation tricks (PPML, distance IV etc.)

Distance elasticity of trade flow: Evidence from Germany

- ▶ RRH (US?)
 - ▶ Log-log:
 - ▶ $[(1 - \sigma)\Phi] \approx -1.5$
 - ▶ $\Rightarrow \Phi \approx 0.375$
- ▶ Ahlfeldt, Roth, Seidel, 2023
 - ▶ Log-log:
 - ▶ $[(1 - \sigma)\Phi] \approx -3$
 - ▶ $\Rightarrow \Phi \approx 0.75$
 - ▶ Log-lin:
 - ▶ $[(1 - \sigma)\varphi] \approx 0.01$
 - ▶ $\varphi \approx 0.0025$



(a) Log-log



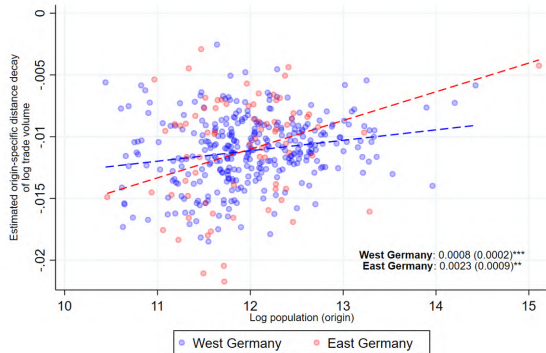
(b) Semi-log

Note: Units of observation are county-county pairs. All variables are residualised in regressions against origin fixed effects, destination fixed effects, an indicator for whether counties are in different states as well as an indicator for whether one county is in East Germany and the other in West Germany. Log trade residuals are averaged within bins: 0.005 log point bins in the left panel and 0.5km bins in the right panel. Averages are computed using the origin population of the county-county pair as a weight. The size of the markers reflect the population size of the origin population.

Q: Which form provides a better fit?

Distance elasticity of trade flow: Evidence from Germany

- ▶ RRH assume uniform distance elasticity
 - ▶ ($\Rightarrow$ symmetric trade costs)
- ▶ But there seems to be heterogeneity
 - ▶ Larger cities trade farther
 - ▶ 'Urban bias'
- ▶ (Ahlfeldt, Roth, Seidel, 2023)



Note: Unit of observation is the county level. The figure plots the estimated origin-specific distance elasticities of trade volumes (given by b_l^T in Eq. (A.33)) against log origin population size at the county level separately for East and West Germany. The dashed lines show the linear fit between the two variables. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Inversion

- ▶ To rationalize the observed data, we need to **invert** $\{A_n, H_n\}$.
 - ▶ We observe the endogenous variables $\{L_n, w_n\}$, have parameterized d_{ni} , and set values for all structural parameters
- ▶ **Two general equilibrium Eqs.** (16, 17) and **two unknowns** ($\{A_n, H_n\}$)✓
 - ▶ Isolate H_n in Eq. (17): $H_n = \left(\frac{1}{\xi}\right)^{\frac{\alpha}{(\sigma-1)(1-\alpha)}} w_n^{\frac{(1-2\sigma)\alpha}{1-\alpha}} A_n^{\frac{\alpha}{1-\alpha}} L_n$
 - ▶ **Please check!**
 - ▶ Plug this into GE Eq. (16) and solve for normalized A_i numerically
 - ▶ Have fun writing a **solver** that starts from guessed values of A_i , returns A_n , updates A_i and iterates until values of geo-mean-one A_i converge
 - ▶ Use the A_n to recover H_n up to a constant

Overidentification

- ▶ We sometimes have **more data than we 'need'** $\Rightarrow$ Should we
 - ▶ improve the quantification of the model?
 - ▶ test the model's predictive power (overidentification)?
- ▶ RRH treat **rent** r_n as an unobserved variable, but **we observe it**
 - ▶ Can introduce residential amenities and use rent to recover them
 - ▶ Population mobility condition will feature residential amenities like in Rosen-Roback
 - ▶ Can also recover rent using the population mobility condition and compare to data
 - ▶ Hope for a **strong positive correlation**
- ▶ What is better depends on the context
 - ▶ Do we argue that the model should be used for out-of-sample prediction?
 - ▶ Do we want to use the model to decompose observed variation in data?

Conclusion

Summary

- ▶ **QSMs of Economic Geography** emphasize the trade between regions
 - ▶ Follow in the tradition of New Economic Geography models
 - ▶ More tractable; rationalize data and allow for realistic counterfactuals
 - ▶ Unlike NEG models, they are built to have deterministic equilibria
 - ▶ Can be a feature or a bug
- ▶ Application follows the same basic structure as QSMs of internal city structure
 - ▶ Use data and structure of model to **estimate** parameter values
 - ▶ Use data and structural parameters to **invert** fundamentals
 - ▶ Use mapping from primitives to equilibrium outcomes to **simulate** the model
 - ▶ Conduct counterfactuals

Next week: **Quantitative Economic Geography Counterfactuals**

Literature I

Core readings

- ▶ Redding, S., Rossi-Hansberg, E. (2017): Quantitative Spatial Economics. Annual Review of Economics, 9, 21–58.

Other readings

- ▶ Ahlfeldt, G. M., Roth, D., Seidel, T. (2023): Optimal minimum wages in spatial economies. Working paper.
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- ▶ Eaton, J. and Kortum, S. (2002): Technology, Geography, and Trade. Econometrica, 70(5), 1741-1779

Literature II

- ▶ Head, K., Mayer, T. (2014): Gravity Equations: Workhorse, Toolkit, and Cookbook. In Handbook of International Economics (Vol. 4), pp. 131-195
- ▶ Helpman, E. (1998): The Size of Regions. In Topics in Public Economics: Theoretical and Applied Analysis, eds. David Pines, Efraim Sadka, and Itzhak Zilcha, 33–54. Cambridge: Cambridge University Press.
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- ▶ Simonovska I, Waugh M. (2014): The elasticity of trade: estimates and evidence. Journal of International Economics. 92(1), 34–50